A120588 concise calculus certificate

VERIFIED

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Definition and first terms

$3\,A(x) = 2 + 1\,x + A(x)^2$, with the origin-normalized solution.

Normalization/grammar: `A(x)=1+(1)*T(x); T=x+1*T^2.`

$a(0), \ldots, a(7) = 1, 1, 1, 2, 5, 14, 42, 132.$

Kernel used throughout: $\rho(u) = u\,(1 - 1\,u^1)$, so $x = \rho(T(x))$.

Typogeometry

literal 2-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2 : 1$.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_1) \in \mathbb{Z}^1_{\geq 0} : \sum_{j=1}^{1} j k_j = n - 1\}, \quad K = \sum_{j=1}^{1} k_j,$$

$$a(n) = \frac{1}{n} \sum_{\boldsymbol{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1, \ldots, k_1} 1^{k_1}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_\gamma \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^{1} P_r(n) a(n+r) = 0$, $n \geq 1$, with

$$P_0(n) = -4\,n + 2$$
$$P_1(n) = n + 1$$

Differential equation

$$(2\,x)\,A(x) + \left(-4\,x^2 + x\right) A'(x) = 3\,x.$$

Matrices, telescoper certificate, and checks

$$G = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 1 & 0 & 2 & -1 \\ -1 & 1 & 0 & 2 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 4 & 2 \\ 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Certificate.

$R(n,u) = \dfrac{2\,u - 1}{\rho(u)^0}.$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n,u) H_n(u))$.

Matrix dimensions: 4×4; remainder 1×2, rank 1, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.